Japan: energy security, economic opportunity and partnership priorities

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AI-generated from the data and checked against it; paper baseline as of 12 October 2026.

Summary

- 5 of the 30 placed sectors of Japan sit in the two low-energy-security corners. Its lowest Energy Security percentiles are in upstream gas and upstream oil, where production falls short of consumption and no reserves are recorded.[1]
- Its highest Economic Opportunity percentiles are in downstream nuclear and downstream coal, where it scores higher than 97% and 96% of the other economies scored.[1]
- On the paper's weighting, its top Strategic Index sector among its own sectors is downstream batteries; it ranks 1st on policy activity but only 15th without the policy term.[1,2]
- The Partnership Strength Index rates the United States highest for Japan, as a reliable supplier. The United States is a priority partner with strong ties today, while China, its 1st trade partner, is a strong tie today that the index ranks lower.[1,3]

1. Energy security risks and why

Of the 30 placed sectors of Japan, 5 sit in the two low-energy-security corners: 5 in the Secure corner, which pairs low opportunity with low energy security, and 0 in the Upgrade corner.[1]

Its lowest Energy Security percentiles are in upstream gas, upstream oil and downstream solar: it scores higher than 1%, 4% and 23% of the other 97, 96 and 147 economies scored. Each sits in the Secure corner.[1]

In upstream gas and upstream oil, its energy imports and reserves pull the score down. Gas production minus consumption was -3.19 EJ in 2025, and oil production minus consumption -6.49 EJ in 2025. For both, the figures use production from the U.S. Energy Information Administration's international statistics, as the Energy Institute does not report it separately. No gas or oil reserves are recorded, which the index counts as zero. For context, its largest suppliers in 2024 were Australia (35.3%) and the United States (15.4%) for gas, and the United Arab Emirates (38.4%) and Saudi Arabia (32.0%) for oil.[3,4,5]

In downstream solar, its investment and reserves pull the score down. On growth in investment it scores 0.00 against a median of 0.48, and on solar resource potential it scores higher than 35% of the other economies.[1]

In downstream batteries, a high-priority sector, its Energy Security is higher than 91% of the other 138 economies scored. Investment pushes the score up, while consumption pulls it down: on growth in use per person to 2035, it scores higher than only 1% of the other economies.[1]

2. Economic opportunities and why

17 of its placed sectors sit in the two high-opportunity corners. Its highest Economic Opportunity percentiles are in downstream nuclear, downstream coal and midstream coal: it scores higher than 97%, 96% and 95% of the other 66, 98 and 154 economies scored.[1]

In downstream nuclear, investment and production push the score up; it produced 94.1 TWh in 2025. In downstream coal, consumption and energy access push the score up; its consumption in this sector was 35.4 GJ per person in 2025.[4]

In midstream coal, trade and technology demand push the score up. It held 5.7% of world exports in 2023, with a revealed comparative advantage of 1.95 in 2024; part of the demand measure is filled in with the world average. For context, its largest markets in 2024 were the United Kingdom (22.5%) and India (13.2%).[3,6,7]

Among its high-priority sectors, upstream wind scores higher than 90% of the other 90 economies, lifted by trade and technology demand. Its revealed comparative advantage in these goods was 3.17 in 2024. In downstream batteries, it scores higher than 79% of the other 138 economies, lifted by investment.[1,7]

3. Sectoral priorities and the existing policy posture

On the paper's weighting, the Strategic Index ranks downstream batteries 1st, upstream batteries 2nd, upstream wind 3rd, upstream solar 4th and upstream nuclear 5th among the sectors of Japan. The lead of downstream batteries is 0.012; the sectors after it are almost tied, with neighbours 0.003, 0.001 and 0.004 apart. On policy activity, downstream batteries ranks 1st, but on data-based priority (the Strategic Index without its policy term) only 15th, so its lead rests on policy. With no sector in the Upgrade corner, the paper's rule takes its 3 highest Strategic Index sectors as high-priority.[1,2]

Its allied pick in the network design is upstream nuclear, ahead of the runner-up, midstream batteries, by 0.023. The pick skips upstream sectors without a resource base, which on the data include upstream batteries, upstream wind and upstream solar.[8]

The Climate Action Tracker rates Japan Insufficient (November 2024). The International Energy Agency's Policies and Measures database holds 302 records for it, 198 of them naming a technology; 0 name both the technology and the stage of upstream wind. Ranked within Japan, policy activity places upstream batteries, upstream wind, upstream solar and upstream nuclear in the middle group. Splitting data-based priority and policy activity into thirds, the sectors with top priority and the least policy are upstream electric vehicles and upstream gas. They rank 7th and 9th on data-based priority but 23rd and 24th on policy activity. Those with bottom priority and the most policy are downstream geothermal and downstream green hydrogen.[1,2,9]

Country Insights also lists, as context, Global Trade Alert's industrial-policy measures for Japan.

4. Partnership priorities and why

The Partnership Strength Index scores how much a partner offers as a reliable supplier, an export market and a place to build supply chains together. From the viewpoint of Japan, it rates the United States highest, as a reliable supplier. Its best supplier scores are in midstream batteries, midstream solar, upstream gas, midstream electric vehicles and midstream wind. The United States is also its 2nd trade partner on average across its energy-goods baskets, and both belong to the G7, the Indo-Pacific Economic Framework and the Minerals Security Partnership. Firms from Japan have 30 cross-border factory projects there.[1,3,10,11]

The index then ranks the Czech Republic 2nd, as an export market, and Canada 3rd and Portugal 4th, each as a reliable supplier. The Czech Republic and Portugal are each a priority partner with few ties today: they are only its 32nd and 80th trade partners.[1,3]

Some large trade partners rank lower on the index. China is its 1st trade partner and Malaysia its 4th, but the index ranks them 48th and 39th; each is a strong tie today that the index ranks lower. In votes at the United Nations, Japan and China sit in opposite blocs.[1,3,12]

As a partner to the coalition, Japan ranks 24th of 192 economies. In the allied network design, its largest designed shares of coalition production are 20.5% in midstream coal and 13.5% in midstream nuclear.[1,8]

5. Data notes and caveats

See the Country Insights page (12 October 2026 baseline) for each source.

- Scores compare Japan with other economies in the same sector; they are not measures of risk or profit in absolute terms.
- All index values are the paper's baseline as of 12 October 2026; Strategic Insights recomputes them under other weightings.
- Some figures behind these scores are filled in where the source has no value: with zero where a source lists every holder, otherwise with the world average.
- Strategic Index scores within 0.01 of each other are almost tied: small changes in the data or the weights can swap their order.
- Many allied picks are close calls: in the paper, 37% of them are within 0.01 of the runner-up.
- Factory projects are cross-border solar, wind, battery and electric-vehicle manufacturing projects only, announced projects included, which may never be built; domestic investment is not covered (Cleantech Manufacturing FDI Dataset, edition 2026.10.05, CC BY 4.0; counts by the authors, a joint venture counted under its listed parent).

Sources

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